

PURIX X1 · XRF gold analyzer

# Karat isn't guessed. It's measured.

PURIX X1 measures up to 20 elements in gold in 30–60 seconds, with  $\pm 0.05\%$  accuracy and without damaging the piece. No cutting, no touchstone, no acid.



Accuracy

**$\pm 0.05\%$**

Digital multi-channel analysis

Measurement time

**30–60 s**

Per piece

Analysis range

**0.01 % – 99.99 %**

Au, Ag, Pt, Pd, Rh and  
15 more elements

Effect on the piece

**None**

No cutting, scraping or acid

## Measuring in four steps

**It's that simple.**

Karat, fineness and 20 elements – all with a single tap.

**1**

**Place**

Put the piece on the measuring window in the chamber.

**2**

**Start**

Close the lid and tap START.

**3**

**X-ray and fluorescence**

The tube sends X-rays into the piece from below. Each element answers at its own energy: gold at 9.71 keV, silver at 22.16 keV.

**4**

**Result**

The detector counts these rays and the software turns them into ratios. Karat, fineness and all 20 elements appear on screen in 30–60 seconds.

## Why XRF?

| Criterion           | Touchstone       | Acid test                | PURIX XRF                     |
|---------------------|------------------|--------------------------|-------------------------------|
| Effect on the piece | Leaves a scratch | Etches the surface       | None                          |
| Result              | Karat by eye     | Karat range (e.g. 14–18) | Percentage of all 20 elements |
| Time                | 1–2 minutes      | 2–5 minutes              | 30–60 seconds                 |
| Plating detection   | No               | Limited                  | Reveals thin plating          |

## Specifications

|                                                                                                                      |                                                    |
|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| Detector<br><b>Customized Si-PIN</b>                                                                                 | Analysis<br><b>Digital multi-channel</b>           |
| Accuracy<br><b>±0.05 %</b>                                                                                           | Analysis range<br><b>0.01 % – 99.99 %</b>          |
| Measurement time<br><b>30–60 s</b>                                                                                   | Sample type<br><b>Solid, liquid, powder</b>        |
| Display<br><b>12" IPS capacitive touchscreen</b>                                                                     | Power supply<br><b>AC 220 V, 50 Hz</b>             |
| Rated power<br><b>&lt;150 W</b>                                                                                      | X-ray tube<br><b>50 kV / 1 mA</b>                  |
| Tube target and focus<br><b>Tungsten, <math>\phi</math>0.5 mm</b>                                                    | High-voltage supply<br><b>50 kV / 1 mA digital</b> |
| Collimator<br><b><math>\phi</math>2.5 mm</b>                                                                         | Noise<br><b>≤50 dB</b>                             |
| Operating temperature<br><b>5–30 °C</b>                                                                              | Humidity<br><b>15–85 % (non-condensing)</b>        |
| Protections<br>X-ray tube overheating protection · High-voltage protection · Warm-up and anti-interference procedure |                                                    |
| Measurable elements<br>Au, Ag, Pt, Pd, Re, Ir, W, Cd, Cu, Ni, Zn, Rh, Ru, Fe, Co, Os, Pb, Sn, In, Mn                 |                                                    |

All values are measured under the manufacturer's test conditions and may vary with the shape and surface of the sample.

## Who it is for

### Jewellers

Measure karat in front of the customer when buying scrap, and back your price with numbers.

### Pawnbrokers and gold lenders

Test every piece taken as collateral; spot thin plating at the counter.

### Bullion and coin dealers

Check the surface composition of bars and investment coins in seconds, without damage.

### Refineries

Verify incoming material and outgoing bars on the same instrument, and track impurities.

### Wholesalers and workshops

Verify alloy recipes before and after production; measure palladium in white gold.

### Assay offices and laboratories

Check karat.

## Service and support

### Installation & training

Installation, staff training and first calibration support; scope stated in the quote.

### 1-year warranty

1-year manufacturer warranty; service support after the warranty.

### Reference sample

A reference calibration sample is supplied; additional karat sets on request.



### Body

Porcelain-white body, graphite-black lid, champagne-gold display frame

### Lid

Brushed gold handle; gas struts keep it open and let it close quietly

### Ventilation and status

Perforated side ventilation grille; vertical gold status light beside the display

## Request a quote or an on-site demo

Scan the QR code or fill in the quote form on our website; we reply as soon as possible.

Web  
[www.purixxrf.com](http://www.purixxrf.com)

Quote form  
[purixxrf.com/#teklif](https://purixxrf.com/#teklif)

Instagram  
[@purixxrf](https://www.instagram.com/purixxrf)

